

FIT9132 - Database Assignment 1

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Reasons for Surrogate Keys

I have added surrogate keys for BOOKING and ITINERARY Entities.

The BOOKING entity had cruise_id, cabin_no & pass_id as Primary Key. Since there shouldn't be an entity with more than two attributes as key, I have added Surrogate key to BOOKING.

This makes a new Primary Key called booking_no as a Primary Key for BOOKING Entity and cruise_id, cabin_id & pass_id becomes Unique Foreign Keys pointing to CRUISE, CABIN and PASSENGER Entity respectively.

The ITINERARY entity had date_time as Primary Key, cruise_id & port_code as a Primary Foreign Key. Since date_time, cruise_id & port_code together is a Primary Key for ITINERARY, I introduced a Surrogate key. This makes a new Primary Key called itinerary_no as a Primary Key for ITINERARY Entity and date_time becomes a Unique Attribute while cruise_id & port_code becomes Unique Foreign Keys.

Assumed Constraints

I have added a total of 3 Check Constraints for my Logical Model.

1. Check Cabin Class

This constraint is to check the different types of cabins available in the CABIN Entity. As per the question, we are given 4 different Cabin Class namely Interior, Ocean View, Balcony and Suite.

2. Check Passenger Gender

This is not given in the question but I have added a Constraint to pass_gender in PASSENGER Entity to check for passenger gender whether it is Male or Female.

3. Check Passenger Phone Number

This constraint is to check whether the passenger is a minor (Below 18) and if they are minor, the guardians phone number should be placed in this passenger's phone number field.